Chapter 4: Natural Language Processing

Natural Language Processing (NLP) focuses on enabling computers to work with human language. NLP tasks include text classification, sentiment analysis, translation, summarization, question answering, and information extraction. Before modern neural methods, many systems relied heavily on manually designed linguistic features.

Modern NLP commonly represents text using numerical vectors called embeddings. Embeddings place words, sentences, or documents into a vector space where semantically related content can be close together. Transformer models have become important because attention mechanisms allow models to consider relationships between different parts of a sequence.